

THE STRUCTURE OF MATHEMATICAL REALITY

Phase IV — Chapter 14

What a Model Is and Why It Works

Assignment

Self-Study Curriculum

Return to the assignment from Chapter 6, where you wrote a structured argument about what happens when a model makes an accurate prediction. Revise it in light of what you now know. Your revision should incorporate:

- The concept of a model as a structure-preserving map
- The three modes of failure
- The connection between model simplicity and predictive strength
- The distinction between statistical compression and generative mechanism capture

Compare the two versions. Write a half-page reflection on what changed in your understanding.